SMART INDIA HACKATHON 2026 — TECHNICAL DOSSIER
AICTE PROBLEM STATEMENT PS 26263

# APAARSetu (■■■■ ■■■■)

APAAR ID-Based Student Data Management & Verification Platform
All India Council for Technical Education (AICTE) — Regulation Bureau

## 1. Executive Summary & Problem Scope

Under the National Education Policy (NEP 2020), the Ministry of Education instituted the Automated Permanent Academic Account Registry (APAAR) — a 12-digit lifelong digital identity assigned to students across India. For the All India Council for Technical Education (AICTE), regulating over 10,000 approved technical institutions requires real-time reconciliation of student enrolments against approved annual intake limits across engineering, management, and polytechnic programmes.

- **Intake Quota Enforcement:** Real-time detection of excess admissions exceeding approved branch quotas.
- **Supernumerary Quota Monitoring:** Tracking NRI, Foreign National (FN), OCI, and PIO candidate allotments.
- **Zero-Knowledge Collision Detection:** Identifying cross-institutional duplicate enrolments without exposing plaintext student tokens.
- **Controlled Academic Verification:** Standardized verification workflow for external agencies with verifiable student consent.
- **Non-Privileged Institution Onboarding:** Strict PENDING_VERIFICATION gating requiring Central AICTE approval prior to data ingestion.

## 2. System Architecture & Core Stack

| Component Layer | Technology Specification | Function & Security Control |
|---|---|---|
| **Frontend UI** | React 18.3.1 + Vite (Pure JS) | Government portal design, bilingual (EN/HI), WCAG 2.1 AA. |
| **Backend API** | Python 3.13 + FastAPI (ASGI) | Stateless REST API, OpenAPI docs, strict Pydantic schemas, RBAC. |
| **Database** | PostgreSQL 15 via Supabase | AWS ap-south-1 pooler (port 6543), additive SQL migrations. |
| **Authentication** | Supabase Auth Authority | HS256/RS256 JWT signature verification, session restoration. |
| **Cryptographic Vault** | AES-256-GCM + HMAC-SHA-256 | Field-level encryption at rest; blind index candidate search. |

## 3. Security Subsystem & Defense-in-Depth

**1. Field-Level AES-256-GCM Encryption:** Every APAAR ID is vaulted using authenticated encryption with a unique 96-bit random nonce and institutional AAD binding to prevent cross-institution ciphertext manipulation.

**2. Privacy-Preserving HMAC-SHA-256 Blind Indexing:** Enables national duplicate candidate detection without decrypting student records, preserving absolute student privacy.

**3. Server-Enforced RBAC:** Strict role verification for Central AICTE roles (AICTE_ADMIN, AICTE_ANALYST, AUDITOR) and Institutional roles (INSTITUTE_ADMIN, INSTITUTE_OPERATOR) with horizontal multi-tenant isolation.

**4. Non-Privileged Onboarding:** Newly registered institutions cannot upload records or trigger verification until affirmatively approved by an AICTE Central Director.

**5. CSV Formula Injection Defense (CWE-1236):** Neutralizes spreadsheet code execution by prefixing dangerous characters with an apostrophe.

## 4. Verification & Testing Evidence

- **Automated Pytest Suite:** 38 tests passed, 0 failures (100% pass rate).
- **Live Supabase Database Integration:** Health check, self-registration, isolation barrier, central approval, transfers, and password recovery fully verified.
- **Frontend Production Build:** 46 modules transformed cleanly in 3.86s, 0 syntax errors, pure JavaScript/JSX.